FEI HAN

Robot Learning · Dynamics Modeling · Bioinspired Robotics

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Joint PhD Program · Expected completion June 2027

Website QR

Education

Westlake University and Zhejiang University
Joint PhD Program · Computer Science and Technology
Aug 2023 - Jun 2027 (expected) · Advisor: Prof. Dixia Fan · i4FSI Lab

University of Chinese Academy of Sciences · MEng · Sep 2019 - Jun 2022
Pattern Recognition and Intelligent Systems · Advisor: Xingang Zhao
Research at the State Key Laboratory of Robotics, Shenyang Institute of Automation, CAS.

Zhengzhou University · BEng
Measurement and Control Technology and Instruments · Sep 2015 - Jul 2019

Doctoral Research Project

Bioinspired Propulsion and Robot Learning

Dec 2023 - Present · Westlake University · Doctoral research

Pursue parallel research on modeling and learning-based control of bioinspired aquatic actuators, and quadrupedal swimming control.

Bioinspired aquatic actuator modeling and control: Conduct force experiments, hydrodynamic modeling and learning-based control for webbed feet, fins and tails; contribute to REEF's real-fluid learning framework and experimental platform, and HydroWrench's offline load reconstruction benchmark.

Quadruped swimming control and safe learning: Use component hydrodynamic models for whole-robot gait optimization and hardware validation; transfer periodic gaits learned safely on a single leg to the robot. Study behavior planning and route tracking in simulation.

Selected Project Outputs

Learn to Swim · **ICRA** 2025 · Published · **First author**

Swimming Under Constraints · **ICRA** 2026 · Accepted · **Co-first and co-corresponding author**

REEF / Embracing Flow Unsteadiness · **Nature Mechanical Engineering** · Under review · **Co-first author, listed first**

From Gaits to Routes · **ICRA** 2027 · Under review · **First author**

HydroWrench · **ICLR** 2027 · Under review · **First author**

Applied Collaboration

DEEP Robotics Arctic quadruped: Contributed a force-testing, modeling and motion-parameter-optimization framework adopted by the collaboration, and joined specialized foot-design discussions. Application: Arctic expedition, Aug 2025.

Research and Engineering Experience

Multilegged Swimming and Bioinspired Locomotion

Apr 2026 - Present · Ongoing collaboration · Manuscript in preparation

Visiting Scholar at Pennsylvania State University, Apr - Oct 2026. **Lead research on multilegged swimming**, including reduced-order dynamics modeling, numerical simulation and experimental design.

Collaborate with researchers from Harvard University, Carnegie Mellon University and other institutions; the manuscript is in preparation. | Baxi Chong group

Storm-Petrel Locomotion

Jul 2022 - Oct 2023 · Westlake University

Contributed to air-water modeling, deep RL tuning and simulation analysis of storm-petrel pattering and sea-anchoring. Co-first-author paper in **Bioinspiration & Biomimetics**, 2023. Research Assistant: Jul 2022 - Aug 2023; continued into early PhD study.

Deep-Sea Soft Robotic Gripper

Sep 2019 - Jun 2022 · MEng research · Core student contributor

Contributed to NSFC Major Research Plan Integrated Project No. 92048302: Fundamental Research on Robots Combining Rigid, Flexible and Soft Structures for Operations in Extreme Environments.

Led structural design, fabrication, sensing and control development. Discussed technical approaches with the advisor and guided junior students in implementation and experiments; their later projects built on the structure and sensing work. Contributed to background research and preparation of project proposal materials.

Zhengzhou University RoboMaster Team

Sep 2016 - 2019 · **Captain: 2017-2018 academic year**

Participated in the 2017, 2018 and 2019 seasons. Developed STM32-based hardware and C control software for infantry, hero, engineer, aerial and sentry robots, including chassis, gimbal and projectile-feeding controls.

As captain, coordinated technical planning, task allocation, subsystem integration and development schedules while remaining involved in implementation and troubleshooting.

Methods and Tools

Learning and modeling: temporal models, constrained RL, imitation learning, policy distillation and multi-objective optimization.
Implementation: Python, MATLAB, MuJoCo, ROS, STM32/C; force sensing, synchronized acquisition and robot integration.

Research Groups

i4FSI Lab · i4fsi.lab.westlake.edu.cn

Baxi Chong group · sites.psu.edu/libr/